

SDF_Project2_Report

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Feature Added

- Introduced `try{} with{} block` for structured exception handling.
- Added two exceptions:
 - **DivisionByZero**: Raised when division by zero occurs.
 - **GenericException**: Raised when an undefined operation is performed (e.g., adding two values of different types).

Changes Made

1. Lexer

- Added new tokens:
 - `try`
 - `with`
 - `{`
 - `}`
 - `DivisionByZero`
 - `GenericException`

2. Syntax

- Added a new AST node: `TryWith()`.

3. Parser

- Added grammar rules to handle the `try{} with{} block`.

4. Type Checking

- Added type checking for `TryWith()`.
- We only check the type of the expression in the with block if there is an error in the try block expression; otherwise, the machine handles it.

5. Machine

- Introduced a new machine value:
 - `MException` of type `string` to hold the exception name.
- Added a new instruction:
 - `IHandle` to handle the `try{}` `with{}` block.
- When an exception occurs:
 - An `MException` with the exception name is pushed onto the machine stack when we encounter some behavior that is undefined.

Correct Cases

- `try {42} with { | DivisionByZero -> 0 };;`
- `try {3 * 5 + (if 5 < 6 then 10 else 100)} with { | DivisionByZero -> 0 };;`
- `try {21/6} with { | DivisionByZero -> 0 };;`
- `try {21/0} with { | DivisionByZero -> 0 };;`
- `let multiplyby = fun f (n : int) : int is if n = 0 then 0 else 2*n;;`
`try {multiplyby 5} with { | DivisionByZero -> 0 };;`
- `let fact = fun f (n : int) : int is if n = 0 then 1 else n * f (n-1);;`
`try {fact 5} with { | DivisionByZero -> 0 };;`
- `let primedividebyn = fun f (n : int) : int is 333/n;;`
`try {primedividebyn 5} with { | DivisionByZero -> 0 };;`
- `let primedividebyn = fun f (n : int) : int is 333/n;;`
`try {primedividebyn 0} with { | DivisionByZero -> 100 };;`
- `let safeprimedividebyn = fun f (n : int) : int is if n = 0 then 0 else 333/n;;`
`try {safeprimedividebyn 0} with { | DivisionByZero -> 100 };;`
- `let safeprimedividebyn = fun f (n : int) : int is if n = 0 then 0 else 333/n;;`
`try {safeprimedividebyn true} with { | GenericException -> 100 };;`
- `try {2 + true} with { | GenericException -> 100 };;`

Malfunction Test Cases

- `let safe_primedividebyn = fun f (n : int) : int is if n = 0 then 0 else 333/n;;`
`try {safe_primedividebyn true} with { | DivisionByZero -> 100 };;`
- `let safeprimedividebyn = fun f (n : int) : int is if n = 0 then 0 else 333/n;;`
`try {safeprimedividebyn true} with { | GenericException i -> i };;`
- `let multiplyby2 = fun f (n : int) : int is if n = 0 then 0 else 2*n;;`
`try {multiplyby2 \5} with { | DivisionByZero -> 0 };;`
- `try {2 + true} with { | GenericException -> 100, DivisionByZero -> -100 };;`

Commit History

exceptions_new	All users	All time
Commits on May 2, 2025		
fix : small change Aryan Agarwal authored and Aryan Agarwal committed 19 minutes ago		08787e4
feat : type_check.ml Aryan Agarwal authored and Aryan Agarwal committed 25 minutes ago		42b3fdf
feat : Changes in syntax.ml Aryan Agarwal authored and Aryan Agarwal committed 46 minutes ago		db52a3f
Commits on May 1, 2025		
feat : Adding the parser tokens Aryan Agarwal authored and Aryan Agarwal committed 2 days ago		a363971
Commits on Apr 29, 2025		
Adding the rules in lexer Aryan Agarwal authored and Aryan Agarwal committed 3 days ago		48f8d46
Commits on Apr 28, 2025		
refactor: deleted irrelevant folders hghit-harshit committed 4 days ago		f832378
Commits on Apr 24, 2023		
Allow parsers access to runtime environment. andreibauer committed on Apr 24, 2023		43cd5b4

Aryan Agarwal ? master	All users	All time
Commits on May 2, 2025		
Merge branch 'master' of https://github.com/AryanAg2701/SDF_Project_2 hghit-harshit committed 6 minutes ago		3305329
feat : added printing for exceptions hghit-harshit committed 6 minutes ago		babf03d
fix : small change Aryan Agarwal authored and Aryan Agarwal committed 12 minutes ago		3371c50
feat : type_check.ml Aryan Agarwal authored and Aryan Agarwal committed 15 minutes ago		190ab41
feat : Changes in syntax.ml Aryan Agarwal authored and Aryan Agarwal committed 15 minutes ago		a77c337
feat : Adding the parser tokens Aryan Agarwal authored and Aryan Agarwal committed 15 minutes ago		3dc6dd7
Adding the rules in lexer Aryan Agarwal authored and Aryan Agarwal committed 15 minutes ago		fa9bc13
Commits on May 1, 2025		
feat : added case to handle printing exceptions hghit-harshit committed 2 days ago		74b38f6
Commits on Apr 30, 2025		

exceptions

All usersAll time

Commits on May 1, 2025

feat : added case to handle printing exceptions

hghit-harshit committed 2 days ago

74b38f6

<>

Commits on Apr 30, 2025

feat : Added compile code to handle try with block

hghit-harshit committed 2 days ago

b2e6637

<>

feat : Added machine code for handling DivisinoByZero and GenericException

hghit-harshit committed 2 days ago

e227524

<>

Commits on Apr 28, 2025

refactor: deleted irrelevant folders

hghit-harshit committed 4 days ago

f832378

<>

Commits on Apr 24, 2023

Allow parsers access to runtime environment.

andrejbauer committed on Apr 24, 2023

43cd5b4

<>

Commits on Mar 22, 2023

Key Learnings

The key things we learned from this project are, first of all, how a lexer and parser work. Coincidentally, we were taking Theory of Computation this semester, so it was great to see regular languages in the lexer and context-free grammars in the parser. We also learned a lot about collaborative development using GitHub, and improved our workflow with feature branches, pull requests, and merge-conflict resolution.